

CausalCache: Restoration-Guided Event Selection for Long-Horizon GUI Memory

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Abstract

Long-horizon GUI agents must decide which historical visual evidence to expose to an action policy under a limited multimodal context budget. Existing memory controllers commonly learn from recency, similarity, attention, or salience labels, none of which directly measures whether high-fidelity evidence changes the policy’s executable decision. We introduce CAUSALCACHE, a policy-grounded supervision framework in which every memory contains the same deterministically serialized event summaries, while a high-fidelity restoration adds exactly one archived post-action image. We measure how much this visual addition restores a frozen policy’s stable full-history self-behavior. We derive an event-level independent target from coalition-conditioned restoration marginals, then distill the expensive offline values into a query-conditioned fixed-budget gate. Exact subset search and coalition interactions remain offline oracles and diagnostics rather than assumptions made by the deployed selector. The versioned reference is screened for exact action-contract parsing, finite teacher-forced logits, and repeat-stable canonical output; expert top-1 alignment is reported only as a diagnostic and never used to admit or remove states. We evaluate whether restoration supervision improves the closed-loop success–memory frontier and whether retained restoration value predicts success after controlling for next-action likelihood, context cost, and task horizon. **[Pending: The frozen set-conditioned student failed its development criterion and remains a negative ablation. The independent selector is unchanged; insert no confirm or closed-loop method-effect claim until the untouched-output confirm and paired controller study pass their preregistered gates.]**

Introduction

A GUI trajectory can contain screenshots, actions, text arguments, coordinates, and state changes accumulated over dozens of decisions. Passing every event to a multimodal policy is expensive and can amplify context rot, while replacing all screenshots with summaries discards exact visual evidence. A memory controller must therefore decide which historical events deserve high-fidelity exposure at each decision.

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Prior controllers select memory using recency, retrieval similarity, attention, learned salience, or task-state heuristics. These signals need not agree with the frozen action policy’s actual dependence on an event. Two memories can even have similar successor-action negative log-likelihood (NLL) while supporting different closed-loop outcomes. This motivates a narrower question: *how much does adding one event’s archived post-action image to an otherwise unchanged strong low-fidelity history restore the frozen policy’s stable full-history action behavior?*

CAUSALCACHE answers this question offline and uses the answers as supervision. It maintains a raw event archive, a cheap summary/index, and a limited policy-visible high-fidelity context. For preregistered stable-reference states, it evaluates mixed-fidelity coalitions and measures each event’s conditional marginal behavioral restoration. All events retain byte-identical low-fidelity summaries in every coalition; selecting an event adds only its post-action image. We project those interacting marginals into one query-dependent target per event by averaging its empty- and singleton-conditioned gains. The claimed contribution is the GUI intervention surface, the policy-behavior value, its distillation into a fixed-budget independent selector, and closed-loop validation—not a new generic attribution or combinatorial search algorithm. A lightweight query-time gate predicts these values from signed-hash query and event features, their element-wise interaction, and fixed geometry metadata.

Our evaluation is organized around four falsifiable questions:

1. Does a restoration oracle improve executable task success at a fixed policy-visible visual budget?
2. Can a distilled gate approach the oracle without material online latency?
3. Does restoration supervision outperform the same gate architecture trained on next-action NLL or shuffled labels?
4. After matching or controlling for NLL, tokens, and horizon, does retained restoration value still explain terminal success?

Related Work

GUI memory. MementoGUI learns multimodal write, compression, and retrieval operators for a frozen GUI ac-

tion model (Zeng et al. 2026). AndroTMem retrieves anchored intermediate states connected to the current subgoal (Shi et al. 2026). These systems establish the importance of learned controllers and dependency-critical events. CAUSAL-CACHE instead derives selector supervision from the marginal behavior change induced by restoring the high-fidelity version of the same event.

Interventional memory selection. CMI compares agent behavior under memory interventions (Srivastava 2026), while ATMem estimates how active memory states affect GUI task completion (Liu et al. 2026). Our scope is intentionally narrower: we hold the action policy fixed, intervene on low- versus high-fidelity GUI evidence, derive selector supervision from coalition-conditioned behavior changes, and validate the dense decision-time surrogate through executable closed-loop outcomes.

Coalition attribution and subset selection. Permutation marginal contributions originate in cooperative game theory (Shapley 1953). We instead use a frozen, explicitly defined projection of empty- and singleton-conditioned marginals for the deployed independent selector. Full coalition tables, interaction statistics, and exact subset search serve as offline supervision analyses and upper bounds.

Problem Formulation

Let x_t be the current observation at decision t , and let its history be

$$H_t = \{e_1, \dots, e_{t-1}\}, \quad e_j = (x_j, a_j, x_{j+1}, \delta_j), \quad (1)$$

where δ_j is an optional deterministic UI delta. Every event has an archived high-fidelity visual increment and a deterministic low-fidelity record $\tilde{e}_j$. We exclude a redundant candidate when its post-state is already the current observation:

$$C_t = \{j < t : \text{post}(e_j) \neq x_t\}. \quad (2)$$

A query-time selector chooses which candidate images to expose:

$$M_t \subseteq C_t, \quad \sum_{j \in M_t} c_j \leq B, \quad (3)$$

where c_j is the visual-token cost and B is the policy-visible context budget. Every event, including events outside C_t , contributes the same low-fidelity record in every memory. Selecting $j \in C_t$ adds its post-action image and nothing else. Raw events are retained in an external archive, so an event can be retrieved again at a later query; B is not a persistent storage or eviction budget.

The action policy π_0 is frozen. Only the selector is trained. Our primary goal is to improve executable task success at the frozen two-event visual budget without attributing gains to extra visual tokens or a stronger action policy. Other budgets require separately declared evaluations.

CausalCache

Two-Level Event Representation

The deterministic low-fidelity record contains

```
step_id, action_type, action_argument,
foreground_app,
screen_text_added, screen_text_removed,
screen_change,
executor_result.
```

The record is present for every event in every coalition and its serialization is byte-identical between low- and high-fidelity conditions. Accessibility content, or a pinned OCR fallback, supplies the normalized screen-text delta. For a selected event, the policy-visible high-fidelity increment is exactly one post-action image: no before image, additional action text, or raw-argument text is added. The raw archive may retain both screenshots and execution metadata, but those fields are not part of the restoration intervention. We report the actual policy-visible text tokens rather than assuming exactly fixed text cost. The frozen gate input is the 200-dimensional vector $[q_{64}, h_{64}, q_{64} \odot h_{64}, g_8]$. Here q_{64} hashes the instruction and current OCR tokens, h_{64} hashes the event summary and candidate OCR tokens, and g_8 contains bounded age, action, OCR, change, and result geometry. Policy-vision features and the requested budget are not gate inputs. For live AndroidWorld histories, an oldest-to-newest adapter rebuilds the same record from the canonical executed action, pinned OCR records, screen-text delta, and 256×256 RGB difference. It assigns age as history length minus one minus chronological index, excludes only the newest current-equivalent post-state, and scores every older event. Fields unavailable in the offline source (foreground app and executor result) remain the same fixed `unknown` values rather than introducing a train-test feature shift. The action policy receives mixed-fidelity inputs through a normal forward pass; we do not splice cached keys and values across incompatible contexts.

Stable Full-History Self-Behavior Reference

Let $X_t(S)$ denote the shared low-fidelity history with post-action images added only for $S \subseteq C_t$. The frozen-policy reference is

$$q_t = \pi_0(\cdot \mid g, x_t, X_t(C_t)). \quad (4)$$

Its deterministically decoded output must be one exact native MobileUse tool call and is used as the canonical action a_t^* . The substrate requires contract-parseable output, finite teacher-forced logits, and identical canonical actions across two deterministic forwards. Expert agreement does not define or filter the reference. Once the untouched confirm split is opened, parse or stability failures count against the fixed-denominator gate rather than being removed. We use the checkpoint’s official tool-template path rather than reproducing its schema in a hand-written prompt. For teacher forcing, the processor’s native assistant prefix is followed directly by the deterministic MobileUse tool call; no generated action description or fixed Action carrier is inserted. Text arguments must already use Unicode NFKC normalization.

Executable Behavioral Distance

For a coalition S , all events retain their low-fidelity records, and only events in S add images. We measure pathwise divergence along the canonical action rather than claiming a

Algorithm 1 Offline Restoration Labels

Require: Stable-reference state $(g, x_t, H_t, C_t, a_t^*)$

- 1: Include the shared low-fidelity record for every event
 - 2: Run $X_t(C_t)$ once to cache full-distinct-history behavior q_t
 - 3: **for** each required coalition S **do**
 - 4: Run the frozen policy on $X_t(S)$ under teacher forcing
 - 5: Record $D_t(S)$
 - 6: **end for**
 - 7: Compute $G_{j,t}^{\text{ind}}$ and the exact budget- B subset oracle
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full sequence-action KL:

$$D_t(S) = \frac{1}{L} \sum_{l=1}^L \text{KL}(q_{t,l}(\cdot \mid a_{t,<l}^*) \parallel q_{t,l}^S(\cdot \mid a_{t,<l}^*)). \quad (5)$$

The KL is teacher-forced over the full vocabulary from the opening `<tool_call>` through the model-emitted closing tag. Standard model EOS tokens that are suppressed during generation receive the same finite minimum logit mask in both reference and candidate distributions before normalization. Action-type match, canonical coordinate-bin match, normalized coordinate distance, and text-argument match are reported as diagnostics, not mixed into the primary restoration distance.

Restoration Targets for Independent Selection

For event j , its conditional marginal under an already restored coalition S is

$$\Delta_{j,t}(S) = D_t(S) - D_t(S \cup \{j\}). \quad (6)$$

These marginals retain redundancy and complementarity because the frozen policy is rerun on the complete mixed-fidelity coalition. Our deployed student is deliberately simpler: it predicts one query-dependent value per event. The frozen independent target averages the empty-coalition and singleton-conditioned marginals,

$$G_{j,t}^{\text{ind}} = \frac{1}{2} \left[\Delta_{j,t}(\emptyset) + \frac{1}{|C_t| - 1} \sum_{i \in C_t \setminus \{j\}} \Delta_{j,t}(\{i\}) \right]. \quad (7)$$

This is an event-level projection of an interacting set utility, not a claim that events are independent. The regression target is $G_{j,t}^{\text{ind}} / \max(D_t(\emptyset), 10^{-12})$. States at or below the denominator epsilon are excluded only from normalized regression and means; their raw target, sign, ranking, and raw evaluation remain present. We measure the projection’s gap to exact subset search. The set-conditioned student is retained only as a negative ablation after failing the frozen development criterion; it is not rescued by changing its threshold, features, or training data.

Query-Time Independent Gate

The distilled selector predicts

$$s_{j,t} = f_\theta(q_{64,t}, h_{64,j}, q_{64,t} \odot h_{64,j}, g_{8,j,t}) \quad (8)$$

with normalized regression and raw-target ranking supervision. At inference, the selector solves a cost-constrained positive-value selection problem:

$$M_t = \arg \max_{M: \sum_{j \in M} c_j \leq B} \sum_{j \in M} \max(s_{j,t}, 0). \quad (9)$$

For equal-cost event blocks this reduces to selecting up to, rather than exactly, the top $b = \lfloor B/c \rfloor$ events whose scores are strictly positive. The threshold is frozen at zero before confirm access; it is not tuned on confirm or closed-loop outcomes. This permits a null decision when all candidate restorations are predicted to be harmful or useless.

Experimental Protocol

Benchmarks and Policies

AndroidWorld (Rawles et al. 2025) is the sole primary closed-loop benchmark. GUIOdyssey (Lu et al. 2025) or an equivalent long-horizon mobile dataset provides offline attribution and action analysis. We use one primary frozen policy and one transfer backbone. Every comparison shares the same low-fidelity channel, action policy, gate architecture where applicable, and visual-token accounting.

The primary policy is frozen to GUI-Owl-1.5-8B-Instruct (Xu et al. 2026) at model revision `06d5faec`, using `bfloat16` and deterministic decoding through its official tool template. We use the frozen v2.2 eager execution substrate with the v2.1 official-tools mixed-fidelity prompt. The earlier GUI-Owl output-contract and UI-TARS expert-alignment gates remain valid negative results; the versioned interface does not relabel those outputs.

The existing policy did not pass the earlier validated-teacher success gate. Accordingly, the closed-loop estimand is the paired effect of the independent memory controller versus equal-budget recent- $B2$ under the same fixed weak backbone, not the performance of a newly validated teacher. AndroidWorld development uses one fixed train instance for each of 60 templates. Infrastructure failures remain in the intention-to-treat denominator, and the sealed test is opened only after the preregistered paired development gate passes.

Each (partition, task type, task index) is one paired unit. Development uses index 0, while the sealed test uses indices 0–2 for each of 25 templates. We first average the three sealed-test paired effects within a template and then average templates equally. Template bootstrap resamples carry all registered instances of the sampled template; the 75 test instances are never treated as independent clusters. Failed episodes receive zero success under the fixed intention-to-treat rule.

The confirm set is not raw-unseen: its source rows and images were processed while constructing the derived artifact. It is, however, policy-output/restoration-output untouched. It contains one decision-6 state from each of 20 trajectories selected before policy inference in frozen hash order. Every state contains summaries for events 1–5. Events 1–4 are the four visual candidates; event 5 is summary-only because its post-state is the current observation. Consequently the reference contains four historical post-state images plus the current observation, covering every non-current historical event

once without duplicating event 5’s current-equivalent image. The primary deployment budget permits up to two of the four candidates. The frozen confirm report compares against the exact at-most-two subset oracle; the complete coalition rows remain raw evidence, but no post-access exact-two or interaction aggregate is added. No state is filtered or replaced after observing parseability, stability, expert alignment, or restoration sensitivity. Image preprocessing targets 2,560 effective visual tokens per image; the actual image grid, visual tokens, and policy-visible text tokens are recorded for every run.

Reference quality remains separate from selector utility. Where a benchmark provides pre-output expert annotations, action-type, coordinate, and text match are descriptive diagnostics only; they never admit, remove, or replace a confirm state. Closed-loop policy quality is measured by official terminal success under the fixed weak-backbone estimand.

Comparisons

The compact main comparison includes summary only, recent, uniform-random exact expectation, OCR-plus-RGB similarity, frozen-policy vision-embedding similarity, the same gate trained on next-action NLL, the strongest reproducible GUI memory controller, a restoration oracle, and distilled CAUSALCACHE. A random-label shuffle uses the same training states and gate capacity as a negative control.

Success–Memory Frontier

We report closed-loop task success against visual tokens and end-to-end latency, plus success per unit memory/compute. Restoration must improve the Pareto frontier rather than consume additional context. **[Pending: Insert main frontier plot and paired confidence intervals.]**

Matched-NLL Mechanism Test

After each authorized partition’s immutable primary closed-loop report, we use the policy-decision states from both origin episodes without deduplicating across origins. At each state, both memories are evaluated against a canonical action generated twice from their visual union, which contains at most four historical images plus the current observation. This pair-union reference makes the symmetric comparison executable under the five-image policy envelope. We compute mean teacher-forced action-token NLL and raw restoration mass $R_m = D_t(\emptyset) - D_t(M_m)$ for each memory. Aggregation is equal-state within each origin episode, equal-half over the two origin episodes, equal-instance within a template, then equal-template. A template is mechanism-eligible only when every preregistered instance pair and every one of its decision states passes the frozen repeat, finite-metric, selection, and budget audits.

The primary matched caliper is fixed in advance at 0.05 nats per token; 0.02 and 0.10 are sensitivity analyses. Within NLL-matched templates, the directional statistic is $\text{sign}_{10-12}(R_{\text{ind}} - R_{\text{recent}})(Y_{\text{ind}} - Y_{\text{recent}})$, with restoration ties contributing zero. The train-60 analysis is development-only. We make a primary mechanism claim only on sealed-test-75, only if at least 15 matched template clusters remain

and the frozen 90% template-bootstrap lower bound is positive. Reference or metric failures make a template ineligible only for this mechanism analysis; they remain in the closed-loop intention-to-treat denominator. We never choose a caliper from the observed success or NLL distribution. This is a post-treatment mechanism association on controller-conditioned state distributions, not a causal mediation effect.

Attribution Cost and Stability

We vary coalition samples K and report attribution standard error, Spearman rank correlation to a high- K estimate, top-budget Jaccard overlap, oracle success, total offline labeling compute, and online selector latency. The per-permutation telescoping identity is used only as an implementation sanity check. Interaction analyses use the already frozen development studies rather than a statistic invented after confirm access. The confirm experiment uses the exact 16-coalition table for its four candidates. Its primary statistics aggregate raw utility over the fixed 20-state denominator; $D_t(\emptyset) > 10^{-12}$ is used only to count memory-sensitive states. No small-denominator state is removed or promoted to a separate primary average.

Limitations and Claim Boundary

Restoration value is a dense decision-time surrogate. It is not a direct long-horizon causal effect, and its relevance to long-horizon control must be established through closed-loop success and the matched-NLL analysis. The causal language in this paper refers only to a controlled low-/high-fidelity intervention on a frozen policy’s behavior. We do not identify environment-level structural causality and do not learn a world model. The method should be weakened or abandoned if restoration fails to beat recency, similarity, and attention; if its matched-NLL association disappears; if the distilled gate cannot approach the oracle; or if gains come from extra tokens. A parseable and repeat-stable self-behavior reference can still be consistently wrong with respect to an expert or the environment. Restoration therefore measures fidelity to a frozen policy, not action correctness. Expert alignment and executable closed-loop success remain separate diagnostics and validation requirements.

Conclusion

CAUSALCACHeturns mixed-fidelity behavioral restoration into policy-grounded supervision for GUI memory selection. Its central empirical burden is not that attribution can fit a selector, but that budget-aligned restoration identifies high-fidelity evidence that improves executable long-horizon control beyond what next-action likelihood and context size explain.

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